

Solver-Integrated Lossy and Lossless Compression for Scalable Flow Simulations

Viral Sudip Shah

University of Illinois Urbana-Champaign



THE STORAGE AND I/O WALL IN EXASCALE COMPUTING

Modern GPU-accelerated HPC simulations routinely generate datasets ranging from hundreds of gigabytes to **multiple terabytes** per snapshot, making storage, I/O, and data movement major bottlenecks in exascale scientific workflows. As compute capability continues to scale, filesystem bandwidth and storage infrastructure increasingly limit output cadence, visualization, and downstream analysis.

BOTTLENECKS

- Multi-terabyte simulation snapshots**
- Filesystem and MPI-I/O bottlenecks**
- High storage and data transfer costs**
- Limited output cadence at extreme scale**
- Expensive post-hoc analysis workflows**



Result: I/O and storage wall limits science and productivity

OUR SOLUTION: SOLVER-INTEGRATED COMPRESSION

We integrate both **lossless** compression for restart/checkpoint reliability and **error-bounded lossy compression** for analysis-oriented outputs directly into the GPU-enabled spectral-element solver nekRS. By compressing solution fields in situ before MPI-I/O, we substantially reduce I/O volume and storage cost while preserving scientific value.

BENEFITS

- Reduces storage footprint and I/O volume**
- Minimizes data movement overhead**
- Preserves QoIs within controlled tolerances**
- Enables higher output cadence at extreme scale**
- Maintains scientific usability of reconstructed fields**



Result: Lower storage & I/O cost, higher output cadence, and scalable science

METHODOLOGY: COMPRESSION-AWARE I/O FRAMEWORK

A portable, compression-aware I/O framework integrated in CFD solver



HPC Solver (nekRS)

- Spectral-element discretization
- High-order accuracy
- GPU-accelerated



Rank-local Fields (Element-based)

- Each rank holds a subset of elements
- No communication for compression



Compression (in situ)

- Error-bounded SZ3 for analysis
- Lossless Blosc2 for checkpoints



Portable Compressed Format

- Rank-decomposable metadata
- Decouples storage from MPI layout
- Supports arbitrary read-time parallelism



Readers & Workflows

- Visualization · Post-hoc analysis
- ROM · ML / GNN pipelines
- In situ or ex situ access



Rank-local data reduction preserving portability across downstream workflows

QoI PRESERVATION

Guaranteed pointwise error bound $|u_i - \tilde{u}_i| \leq \epsilon_{rel} \cdot \Delta u_{range}$ or ϵ_{abs}
Closed form a priori bounds on physical QoI

Define: $\eta = \epsilon_{rel} \cdot (\Delta u_{range})^{max}$ or ϵ_{abs} , Pick ϵ from target accuracy

QoI	Formula	Error-bound scaling, small η
Mean profiles	$\langle u \rangle$	η
RMS / variance	$\langle u'^2 \rangle = \langle u^2 \rangle - \langle u \rangle^2$	$4\eta \cdot \max(u)$
Velocity magnitude	$ u = \sqrt{u^2 + v^2 + w^2}$	$\sqrt{3} \eta$
Kinetic energy	$E_k = \frac{1}{2} u ^2$	$\sqrt{3} \eta \cdot \max(u , v , w)$
Turbulent KE	$k = \frac{1}{2} u' ^2$	$6\eta \cdot \max(u' , v' , w')$
Gradients vorticity, stress etc.	$\nabla u, \omega = \nabla \times u, \tau = \nabla u + (\nabla u)^T$	$O(\eta/h)$ <i>grid-dependent</i>

WORKED EXAMPLE: FUSION BLANKET VELOCITY

Given: $u_x, u_y, u_z \in [-58, 58]$, $\epsilon_{rel} = 10^{-3}$, $\Delta u_{range} = 116$

Per-component bound: $|u_i - \tilde{u}_i| \leq 0.116$

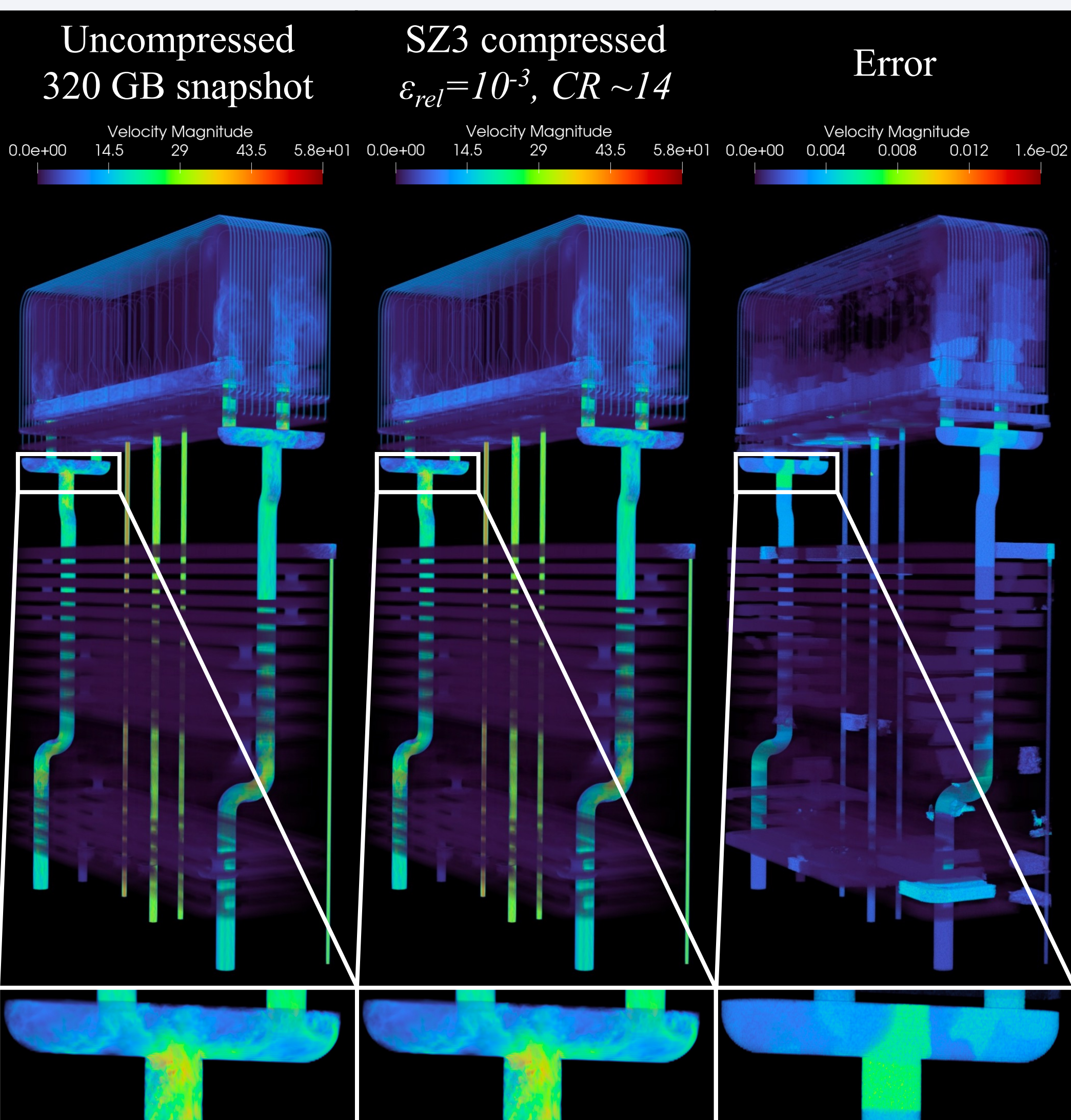
Magnitude bound: $\|u\| - \|\tilde{u}\|_{\infty} \leq \sqrt{3} \cdot \epsilon_{rel} \cdot \Delta u_{range} \approx 0.20$ (worst-case)

Observed: $\|u\| - \|\tilde{u}\|_{\infty} \leq 1.6 \times 10^{-2} \sim 12 \times$ tighter than worst-case



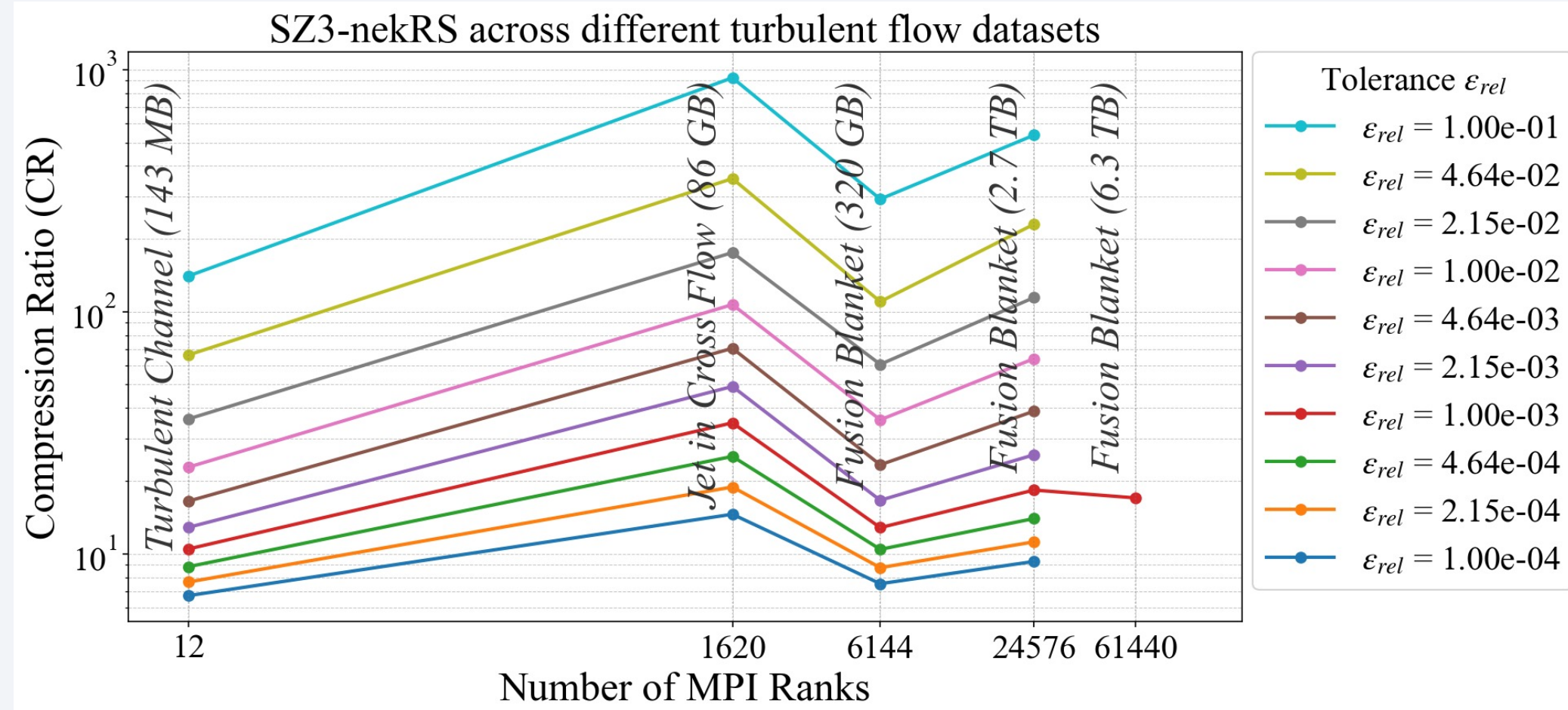
Pointwise tolerance translates to QoI guarantees

FUSION BLANKET: VELOCITY MAGNITUDE

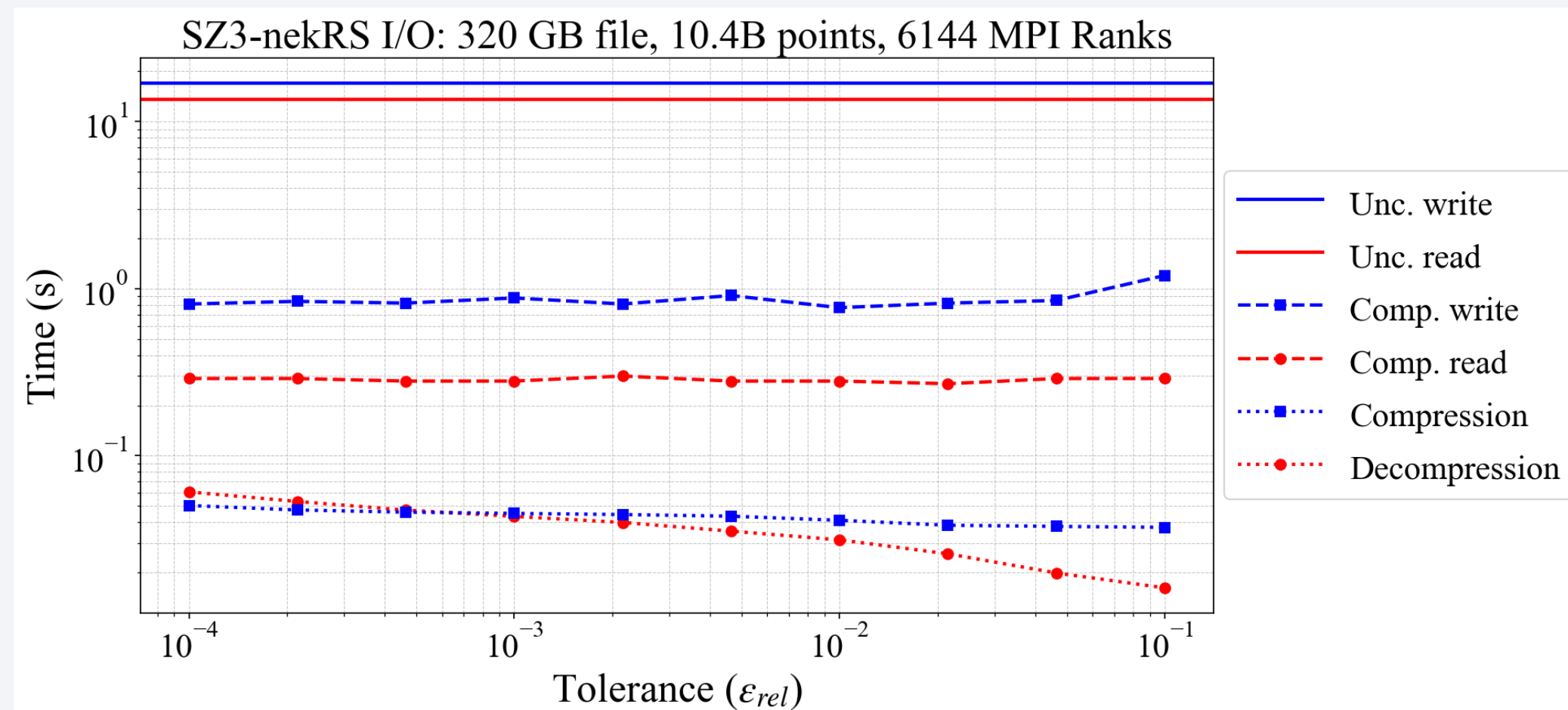


Key flow structures are preserved after error-bounded compression

PERFORMANCE: SCALING & I/O TIME

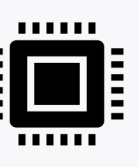


Scaling validated across multiple flow cases beyond 60,000 MPI ranks



10-100x compression ratio and up to 20-50x I/O time reduction

FUTURE WORK



- GPU-resident compression and asynchronous I/O
- SEM-aware transform-based compression strategies



- Integration with visualization tools (ParaView / VisIt)
- Evaluation of additional lossless and lossy compressors

Additional directions:



- Reactive and multiphysics flows
- Adaptive tolerance selection



- Compressed ML-ready datasets for ROM and AI workflows

GENERALIZATION



- Applicable to a wide range of distributed scientific workflows with locally structured data layouts.



- High-order FEM / DG

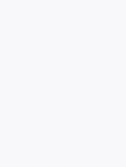


- Structured-grid CFD



- Multiphysics workflows

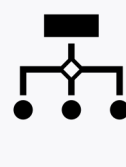
Key ideas:



- Rank-local, in situ data reduction



- QoI-aware error-bounded compression



- Compressed parallel MPI-I/O



A general strategy for scalable, data-centric HPC workflows

ACKNOWLEDGMENTS



Thanks to Yu-Hsiang Lan, Harikrishnan Tummalapalli, Shivam Barwey, Riccardo Balin, Ramesh Balakrishnan, Paul Fischer, Sheng Di, Franck Cappello, and Rupert Eardley-Brunt for discussions and technical support related to nekRS, scientific compression, visualization, and large-scale simulation datasets.



This work used resources of the Argonne Leadership Computing Facility (ALCF) and Oak Ridge Leadership Computing Facility (OLCF), U.S. DOE Office of Science user facilities supported under Contracts DE-AC02-06CH11357 and DE-AC05-00OR22725.



Generative AI tools assisted with language editing and code/debugging tasks.



Data reduction is as critical as numerical computation in the exascale era

REFERENCES & MORE

- V. Shah et al. 2026. Solver-Integrated Lossy and Lossless Compression for Scalable Flow Simulations. PASC '26. <https://doi.org/10.1145/3815572.3815766>
- P. Fischer et al. 2022. NekRS: A GPU-accelerated spectral element Navier-Stokes solver. Parallel Comput. 114, 102982. <https://doi.org/10.1016/j.parco.2022.102982>
- K. Zhao et al. 2021. Optimizing Error-Bounded Lossy Compression for Scientific Data by Dynamic Spline Interpolation. ICDE 2021. <https://www.osti.gov/servlets/purl/1864145>
- Blosc Development Team. Blosc2: A fast compressed data store library. <https://blosc.org>

Scan for a digital copy of the poster

